



Winding Tree

How Blockchain Could End Travel Industry Pain Points — For Real



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Following on [the brilliant article from Trond Vidar Bjørøy](#), I'd like to elaborate on the points Trond outlined in his article. Trond, thank you for your clarifications and comments on this text!

Overbooking

Blockchain's ability to prevent double spending could, in theory, remove the problem of double bookings in the industry, which would hopefully (for both passengers and airlines) eliminate instances like this from happening.

One of the companies that I founded a while ago was Walksource. It aimed to solve the problem of overbooking at hotels and I know for sure that hotels and other travel companies will never stop that practice because it is a model that brings additional revenue to the table. Call ahead to make sure they hold your room for you.

Fraud and Chargebacks

Merchants that sell travel are typically seen as high risk because of the amount of refunds and chargebacks that occur in the industry.

With blockchain, once a payment has been made, typically you can't reverse it. And this mechanism will make fraudulent cases easier to spot and less likely to occur.

Fraud and chargebacks are two sides of the same coin. We call “fraud” a case when there is a transaction that was submitted by a fraudulent party, e.g. someone who is using someone else's credit card. A chargeback, on the other end, is needed when the receiving party is fraudulent.

Chargebacks

Chargeback is a feature, it's not something that we need to eliminate. We do need a new system for having chargebacks available on a platform with irreversible transactions. It already can be done without an intermediary, e.g. by transacting via an escrow smart contract that would only release the funds to the seller if the buyer explicitly allows it. E.g. when you check in at a hotel, you have to provide a secret code to the front desk agent that only you possess, that was generated at the time of booking. Without the code you can't check in, because it acts as a proof of your booking. But when you disclose that code to the hotel, they have proof that the service has been rendered.

In more complicated cases fully decentralized arbitration platforms, like Kleroterion, will be used.

Fraud

There will be less fraud on the blockchain, indeed, but not because the payment can't be reversed, but because the underlying principles of the blockchain technology require its users to have better overall security. Your average crypto owner (if they don't just store their BTC on an exchange because it's convenient) is light years ahead of an average internet dweller that uses the same password on all the sites they register.

When we start paying for goods and services with cryptocurrencies, such as Líf, the fraud problem will be eliminated simply because if someone is able to access your private key and send a transaction on your behalf, then you have a much bigger problem than one fraudulent transaction. The attacker won't spend your BTC/ETH/LIF on booking hotels (a very amusing scenario!), they will simply transfer all your funds to the account that they control, as quickly as possible.

As opposed to crypto, credit card security today looks like a joke. Your one and only key (your CC number and all other information on your credit card) are exposed EVERY TIME you make a transaction. And when you tick the "store payment method for later" that key is stored in a database which will be accessed by at least employees of that company and, in many cases that are simply too hard to ignore, by other parties, usually with malicious intent in mind.

The downside of the new way is that you have to educate yourself about the implications of having one master password, your private key. First of all, if you lose it, this is it, your funds are gone for good, there is no call center to call. Secondly, no one else should be able to gain access to your private key, because if they do, your funds are gone for good. [Andreas M. Antonopoulos](#) explains it very well:

The best practices for having your private keys at hand and yet not lose control of them is a combination of a hardware wallet (like Trezor or Ledger) and cold storage (fancy term for writing your private key or mnemonic phrase down on a piece of paper).

The upside of this system is that we can finally have full and complete control over our funds. How can you start on this steep learning curve? Buy some Bitcoins or ETH or any other currency and store it in a wallet private keys from which only you can control. It's easier than you think.

Identity and Reputation

Identity is overrated. I want to be completely anonymous unless I really want to do something that requires me to disclose some (not all!) of my personal data (e.g. entering a nuclear plant or running for president). Why isn't this a default option everywhere? Why does my phone come with a hundred of pre-installed and activated trackers? Why does Facebook require you to supply your real name to its algorithms? The right to be anonymous on the internet should be engraved into all the protocols we use. But I digress, so let's get back to the case where you have to prove who you are.

Decentralization through crypto-economies will enable billions of people to get access to basic financial services, connecting them with the rest of the world.

The statement above is correct: it is much easier to store and manage your identity via a fully automated system, as opposed to obtaining a passport. But there is a “but”, and one that we can’t ignore. To enable banking for the unbanked and provide identities for the unidentified, we need to give them the following, in this order: 1) food, 2) shelter, 3) electricity, 4) computers, 5) education about public key cryptography. The good news is that you, the reader, are already on level 4, so you are just one step away from taking full ownership of your own financial life. But not identity, not just yet.

Even though there are identity systems in use for authentication today, like federated identity and social login, you can’t easily build on them to pull in data from many sources or use the attributes that you want when you want them — unless you integrate with every party. Decentralized identity systems are here to fix that.

Federated identity systems are, first of all, centralized and therefore prone to attacks, like the [mind-boggling leak of 143M US identities by Equifax](#), so let’s see how a decentralized identity would work.

What does a truly decentralized identity on blockchain mean anyway? It means that you and only you control your data, including read and write access to it. E.g. you are checking in to a flight and the airline requires you to prove that you are the same person that bought the ticket, you need to disclose the same nuggets of your personal data (today: name, DOB, etc., tomorrow: biometric data) that you used when you booked it.

Today you prove your identity by using a paper blockchain called the passport. That blockchain has a certain amount of trust that was supplied to it by the issuing authority, usually a government agency. The authenticity of the information in that distributed database is ensured by the security features that that ID has: biometrics, holograms, invisible ink, etc. So far so

good, this technology allows me to disclose my identity only when I want to and it allows other parties to attach additional information to that identity. E.g. I can have a visa attached to it. Or, if your host's AirBnB account has a verified government-issued ID, their AirBnB rating is, in a sense, attached to that ID as well.

There are three problems with your passport though: 1) you have to carry a physical object with you; 2) the information in my passport is not very secure, e.g. if someone simply obtains my passport number, I'm, potentially, in big trouble; and 3) a trust monopoly is necessary in order to issue the ID, therefore the issuance and maintenance of these IDs are very expensive (think about your average DMV visit), not mentioning the censorship problem (try and become naturalized as a US citizen, ha!).

The first two problems can be solved by governments, but it will take decades before it happens. The solution to the third problem though, by definition, will change everything. How do we remove not just governments but any other entity from this equation?

There are already completely sovereign individuals, like Roma Siri, but before we all agree on standards about how these IDs are created, stored and shared, we will not have a truly decentralized identity systems. A nonpartisan body that would develop standards is necessary in this case. A for-profit company developing a decentralized ID is an utter nonsense.

Many companies trying to build these identity systems at the moment — Sovrin, uPort, Civic, æternity, etc, — the ones focusing on the protocol, not the frontend, will be the likely winners.

Settlement and Loyalty

I grouped these two issues into one section because both issues are concerned with the transfer of value and rules around it. Transfer of value is, of course, the core feature of public permissionless blockchains.

If you are not familiar with the travel industry, the settlement process in it in some cases can take weeks or even months. Worse than that, in the United States, there is only one company that does settlement for airlines. One!

Naturally, the settlement, in the case of blockchain, should happen in a completely autonomous way, without a central party. Same goes for loyalty. Now that you have value transfer readily available for travel companies, they should be able to deploy their own loyalty programs and issue their own loyalty tokens. In fact, this is going to be the core feature of the Winding Tree offering.

Processes Automation

We say “process automation”, we mean “smart contracts”. When we say “smart contracts” we mean “process automation”. Smart contracts are the future of business process automation, another core feature of blockchains.

The main difference of smart contracts from other software that is used today for the purposes outlined above is that smart contracts can transfer value without an intermediary, in a trustless environment. Therefore you should not be seduced by offerings of private blockchains to make your business more effective. “Private blockchain” is a cunning oxymoron created to distract lazy and weak-minded in order to promote business as usual under the flag of innovation.

Removing Silos

Will we see a predominance of permissioned and private blockchains in our industry as the big incumbents try to evolve and maintain their power? We're already good at closed ecosystems, so this could be the next natural step in that direction.

The beauty of this technology is that it can't be usurped. If tomorrow all bitcoins were bought out by one party or, say, by three, five or ten, the network would immediately cease to exist. The main value of blockchains

comes from precisely from the fact that they aren't owned by anyone, their ownership is distributed. A company that builds a private blockchain looks like this. The power of blockchain simply dissolves when attempted to control.

Travel Industry desperately needs a body that would unite all the players in a nondominant way, without even a possibility to create a monopoly in the future. If we fail to do so, we won't see any further innovation in the travel industry. Hence, Winding Tree.

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	verification				contract configuration - Approve minting of new LIFs to the "customers rewards" fund	
Registration / Invalidation	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted			
Verification		-Verifies suppliers or retailers -Invalidates suppliers or retailers -Pays verification fee with LIF to the DAO from its deposit balance			-Validates and verifies verifiers -Invalidates verifiers	
Deals creation			-Creates a deal on behalf of a customer and pays an offer price	-Creates a deal and pays an offer price		
Deals rejection	-Pays tx expenses -Pays a proper amount of LIF from its deposit as a rejection penalty			-Receives a full refund of the deal funds	-Receives "rejection fee" from the supplier	
Deals cancellation	-Receives fullpart of	-Receives part of the	-Receives part of the	-Pays tx expenses	-Receives part of the	



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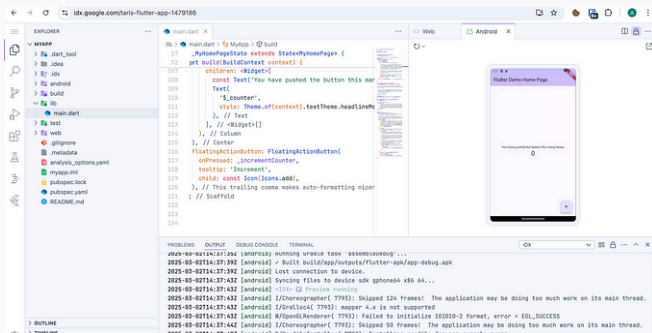
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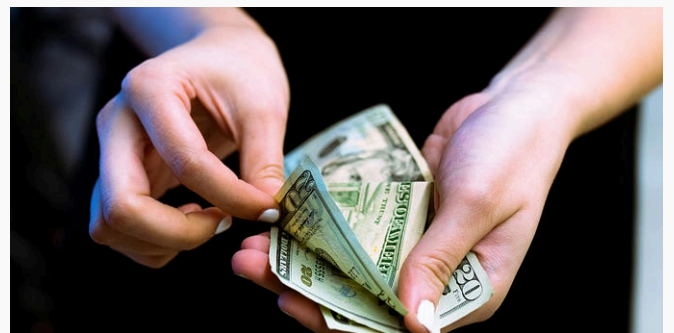
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